Morphisms and Commutative Diagrams

Morphism Types Complete Reference

Monomorphism (Monic, ↪)

f: X → Y is monic if: $f \circ g_1 = f \circ g_2 \Longrightarrow g_1 = g_2$

In **Set**: Injective In **Ring**: $\mathbb{Z} \to \mathbb{Q}$ is monic (non-injective!)

Epimorphism (Epic, ↠)

f: X → Y is epic if: $h_1 \circ f = h_2 \circ f \Longrightarrow h_1 = h_2$

In **Set**: Surjective In **Ring**: $\mathbb{Z} \to \mathbb{Q}$ is epic (non-surjective!)

Isomorphism (Iso, ≅)

∃g: Y → X with g ∘ f = id_X and f ∘ g = id_Y

Endomorphism (Endo)

f: X → X (self-map)

Automorphism (Auto)

Endomorphism that is also isomorphism

Bimorphism

Both monic and epic (but NOT necessarily iso!)

Commutative Diagrams

Diagrams express equalities between morphism paths:

```
     f
 X ───→ Y
 │      │
g│      │h
 │      │
 ↓      ↓
 Z ───→ W
     k
Commutes: k ∘ g = h ∘ f
```

Five Lemma

Classical result for chasing diagrams in exact sequences

```
... → A → B → C → D → E → ...
      |   |   |   |   |
      ↓   ↓   ↓   ↓   ↓
... → A'→ B'→ C'→ D'→ E'→ ...

If outer four preserve exactness, middle does too
```

Snake Lemma

Produces connecting homomorphism between kernels/cokernels

```
0 → A → B → C → 0
    |   |   |
    ↓   ↓   ↓
0 → A'→ B'→ C'→ 0

Gives connecting map: ker(C→C') → coker(A→A')
```

Special Morphism Classes

Split Monomorphism

$f: X \to Y$ with $\exists r: Y \to X$ where $r \circ f = id\_X$ (retraction)

Split Epimorphism

$f: X \to Y$ with $\exists s: Y \to X$ where $f \circ s = id\_Y$ (section)

Regular Mono

Equalizer of some pair of morphisms

Regular Epi

Coequalizer of some pair of morphisms

Normal Mono (in abelian)

Kernel of some morphism

Normal Epi (in abelian)

Cokernel of some morphism

Diagram Chasing Techniques

1. **Element chasing**: In concrete categories, follow elements through diagrams
2. **Freyd-Mitchell embedding**: Abelian categories embed into module categories
3. **Diagram lemmas**: Five lemma, snake lemma, 3×3 lemma

Study Guide

- Master monomorphism vs injectivity
- Understand why $\mathbb{Z} \to \mathbb{Q}$ is monic
- Work through five lemma proof
- Practice snake lemma construction
- Learn diagram chasing in abelian categories

Key Questions

1. Why is monic ≠ injective in general?
2. How do five and snake lemmas relate?
3. What makes diagram chasing valid?
4. How do normal monos differ from regular?